



Shanon Entropy Properties - I

1. **Entropy is Non-negative:** $H(X) = - \sum_x p_X(x) \log p_X(x) = \sum_x p_X(x) \cdot i(x) \geq 0$

- Entropy represents expected information.
- Since $i(x) = -\log p_X(x) \geq 0$, we have $H(X) \geq 0$.

2. **Entropy is Concave:** $H(\lambda p_1 + (1 - \lambda)p_2) \geq \lambda H(p_1) + (1 - \lambda)H(p_2)$

- Entropy increases under the averaging of distributions.
- This reflects increased uncertainty when mixing sources.

3. **Invariance Under Relabeling:**

- Entropy depends only on the probabilities, not on the labels of outcomes.
- Permuting or renaming outcomes does not affect $H(X)$.

Shannon Entropy Properties - II



4. **Zero Entropy for Deterministic Variables:** $H(X) = 0 \iff p_X(x) \in \{0,1\}$ for all x
- If all probability mass is on one value, no uncertainty exists.
 - Learning X yields no new information.
 - **Proof:** $H(X) = 0 \implies p_X(x)\log(p_X(x)) = 0$ for every $x \in \mathcal{X}$.
 - Two possibilities: $p_X(x) = 0$ or $p_X(x) = 1$. Because $\sum_x p_X(x) = 1$, there is exactly one x_0 for which $p_X(x_0) = 1$.
5. **Maximum Value of Entropy:** $H(X) \leq \log |\mathcal{X}|$
- where $|\mathcal{X}|$ is the cardinality of the alphabet
 - Equality if all outcomes are equally likely, that means X is a uniform random variable.

Overview



- Introduction
- Classical Entropy
- **Noiseless Source Coding: Data Compression**
- Noisy Channel Coding: Channel Capacity



Example: Fixed-Length Coding

- **Source:** $X \in \{A, B, C, D\}$ with probabilities:
 $P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{4}, \quad P(C) = \frac{1}{8}, \quad P(D) = \frac{1}{8}$
- **Fixed-length coding:** Use 2 bits per symbol: $A \rightarrow 00, \quad B \rightarrow 01, \quad C \rightarrow 10, \quad D \rightarrow 11$
- **Expected codeword length:** 2 bits
- **Shannon Entropy:**

$$\begin{aligned} H(X) &= - \left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{4} \log_2 \frac{1}{4} + 2 \cdot \frac{1}{8} \log_2 \frac{1}{8} \right) \\ &= \frac{1}{2} + \frac{1}{2} + \frac{3}{4} = \frac{7}{4} = 1.75 \text{ bits} \end{aligned}$$

- *Entropy is lower than the expected length of fixed-length coding.*

Example: Variable-Length Coding



- **Variable length coding:** $A \rightarrow 0$, $B \rightarrow 10$, $C \rightarrow 110$, $D \rightarrow 111$
- **Expected codeword length:** $\frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = \frac{7}{4} = 1.75$ bits
- **Conclusion:**
 - This variable-length code achieves the Shannon limit.
 - Efficient compression: matches $H(X)$.
 - The code is prefix-free implies uniquely decodable.
 - A coding scheme must ensure unique decodability.

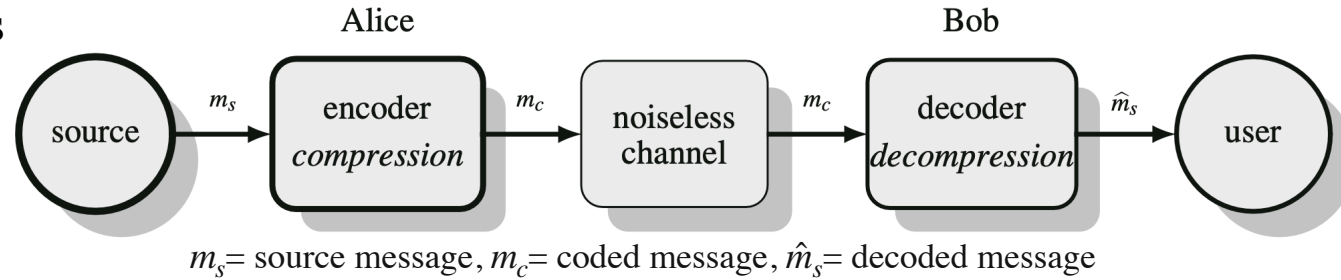


Noiseless Source Coding: Data Compression

- Efficient representation of messages produced by a source of information with redundancy

- Source: a sequence of IID random variables

- $A \sim (\mathcal{A}, p_A)$, where $\mathcal{A} = \{1, \dots, K\}$, word: $A^L := (A_1, \dots, A_L)$, realisations: $a^L := (a_1, \dots, a_L) \in \mathcal{A}^L$ with probability: $P[A^L = a^L] = p_A(a_1) \cdots p_A(a_L)$



- **Theorem:** (Law of large numbers, LLN) In an IID source, the mean estimator: $\mathcal{E}(A^L) := \frac{1}{L} \sum_{n=1}^L A_n$, converges to the mean $m(A) = E[A]$: $\lim_{L \rightarrow \infty} \mathcal{E}(A^L) = m(A)$ (in probability) i.e., for any $\varepsilon > 0$, $P[|\mathcal{E}(A^L) - m(A)| > \varepsilon] \rightarrow 0$ as $L \rightarrow \infty$.
- Entropic Application of LLN: Information of a source word A^L : $i(A^L) = -\log[p_A(A_1) \cdots p_A(A_L)]$ s.t. entropy: $H(A^L) = LH(A)$ and entropy estimator: $\mathcal{J}(A^L) := \frac{1}{L} i(A_1, \dots, A_L) = \frac{1}{L} \sum_{n=1}^L i(A_n) = -\frac{1}{L} \sum_{n=1}^L \log p_A(A_n)$
- **Theorem:** In an IID source, the entropy estimator $\mathcal{J}(A^L)$ converges to entropy: $\lim_{L \rightarrow \infty} \mathcal{J}(A^L) = H(A)$ (in probability)

Shannon's Source Coding Protocol



- **Asymptotic equipartition property:** Set of the possible sequences emitted by source is subdivided into two distinct classes, called *typical sequences* and *atypical sequences*.
- Example: IID binary source $P[A = 1] = p$ and $P[A = 0] = q = 1 - p$. For an L length sequence we expect to see in the sequence approximately Lp 1s, and Lq 0s (Typical Sequence)
- Probability of Typical sequence: $p(a^L) = p_A(a_1) \cdots p_A(a_L) \simeq p^{Lp} q^{Lq}$ and information: $i(a^L) \simeq -Lp \log p - Lq \log q = LH(A)$. Thus, $p(a^L) \simeq 2^{-LH(A)}$
- **Typical Sequences:** (1) are equally likely (uniformly distributed), and (2) their number is at most $2^{LH(A)}$
- **Shannon's Data Compression:** There are at most $N \simeq 2^{LH(A)}$ typical sequences, it is sufficient to create a codebook, or an index set, of cardinality N , where each codeword can be represented by a string of $\log N \simeq LH(A)$ bits, i.e., using $H(A)$ bits/symbol.
- **Definition:** A realization a^L is ε -typical if its sample entropy $\mathcal{J}(a^L) = -\frac{1}{L} \log p_A(a_1) \cdots p_A(a_L) = -\frac{1}{L} \sum_{n=1}^L \log p_A(a_n)$ is ε -close to the entropy $H(A)$, i.e., $H(A) - \varepsilon \leq \mathcal{J}(a^L) \leq H(A) + \varepsilon$.
- The ε -typical set $A_\varepsilon^{(L)}$ is the set of all ε -typical sequences s.t. $2^{-L(H(A)+\varepsilon)} \leq p(a_1) \cdots p(a_L) \leq 2^{-L(H(A)-\varepsilon)}$